

**Operations Training Program
Job Performance Measure**

STATION:	SALEM		
SYSTEM:	Control Air		
TASK:	Perform Actions for a Loss of All AC Power (Start the SBO Compressor)		
TASK NUMBER:	N1150140501		
JPM NUMBER:	11-01 NRC IP-j		
ALTERNATE PATH:	☐	K/A NUMBER:	2.1.23
APPLICABILITY:	IMPORTANCE FACTOR:		
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	In Plant		
REFERENCES:	SC.OP-SO.CA-0001, SBO Diesel Control Air Compressor, Rev. 14 2-EOP-LOPA-1, Loss of All AC Power, Rev. 27 S2.OP-AB.LOOP-1, Loss of Off-Site Power, Rev. 27 S2.OP-AB.CA-0001, Loss of Control Air, Rev. 17 DWG 604495 (Rev. 2) and 205347-2 Rev. 40 (H-6)		
TOOLS AND EQUIPMENT:	SBO Compressor Building is normally unlocked (L-3 key if locked)		
VALIDATED JPM COMPLETION TIME:	12 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	08-13-12
Validated By:	SME or Instructor	Date:	
Approved By:	Training Department	Date:	
Approved By:	Operations Department	Date:	
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:			DATE:

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**TASK
NUMBER:** N1150140501

**INITIAL
CONDITIONS:**

Both Salem Units have tripped due to a loss of off-site power. Equipment problems have resulted in NO Emergency Control Air Compressors (ECACs) running.

INITIATING CUE:

The Unit 2 CRS has directed you to start the Station Blackout Compressor (SBO) IAW SC.OP-SO.CA-0001, SBO Diesel Control Air Compressor, starting with Step 5.1.3.

The Component Off Normal and Off Normal Tagged review has been performed SAT.

It is **NOT** the summer operating period.

The inspection covers have already been removed, Step 5.1.1 was performed SAT, and the inspection covers replaced IAW Step 5.1.2

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Operator performs SC.OP-SO.CA-0001, SBO Diesel Control Air Compressor, SECTION 5.1 in the correct sequence which would result in SBO Compressor running supplying air through 1CA1913 SBO COMPRESSOR DISCHARGE VALVE, and 1CA1886 BLACKOUT AIR COMPRESSOR ISOLATION, and 2CA584 YARD CONTROL AIR SUPPLY VALVE to Aux Building headers 1A and 2A.

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		Provide marked up copy of the SC.OP-SO.CA-0001, SBO Diesel Control Air Compressor with Prerequisites signed off. Additionally, Step 5.1.1 for removing Inspection Covers will be previously completed, as will Step 5.1.2 to re-install inspection covers.	Operator has and uses standard personnel safety equipment. Dons hardhat, safety glasses. Uses hearing protection when entering high noise areas that are posted as hearing protection required. Uses gloves when operating equipment or climbing.		
			Operator reviews and signs Precautions and Limitations, and performs P&L 3.10 as described in next step. Operator <u>must</u> obtain hearing protection per P&L 3.1 All other P&Ls require no action other than reading and initialing with the exception of P&L 3.10 which will be performed at the SBO as identified below.		
			Operator proceeds to SBO Compressor Building.		

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	P&L 3.10	Ensure the Battery Tender has power by observing the Red and Green lamps on the front. The placard next to the Battery Tender explains the Red and Green lamps and their significance.	Operator observes the battery Tender GREEN light is illuminated, and the RED light is not illuminated in the SBO building. (Green is a float charge on the battery, Red (full charging current) would only be present after a discharge)		
	5.1.3	OPEN both Engine intake louvers (located outside building on west wall.)	Cue: After operator locates and describes how to open Engine intake louvers, state that the Engine intake louvers are open.		
	5.1.4	OPEN 1CA1920, BLACKOUT AIR COMPRESSOR DRAIN VALVE, to drain any water accumulation, then CLOSE 1CA1920.	Cue: After operator locates 1CA1920, BLACKOUT AIR COMPRESSOR DRAIN VALVE, and describes how to open, including that the drain cap must be removed, state 1CA1920, BLACKOUT AIR COMPRESSOR DRAIN VALVE, has been cycled and no water came out of pipe. Since the cue would be the same whether or not the drain cap was removed (nothing came out the end of pipe) only comment here for review if pipe cap was not removed.		

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	5.1.5	OPEN 2FZSBO10, BATTERY CHARGER, breaker.	Operator locates 2FZSBO10, BATTERY CHARGER, breaker in Panel 2FZSBO, SBO Compressor Bldg Distribution Panel, and simulates opening. Cue: 2FZSBO10, BATTERY CHARGER, breaker, is open.		
	5.1.6	PLACE SBO Control Air Dryer switch in ON position.	Operator locates SBO Control Air Dryer switch and simulates turning switch to ON position. Cue: SBO Control Air Dryer switch is ON, and Control Air On light is illuminated.		
	5.1.7	RECORD Engine Hourmeter reading on Attachment 1, Section 1.0, and in log book at SBO compressor.	Operator locates Engine Hourmeter, 1XA16993, Tachometer/Hourmeter, inside engine control panel labeled Compressor Noise Emission Control Information, and records current reading in Section 1.0 of Att. 1. Operator locates logbook hanging from side of panel 2FZSBO, SBO Compressor Bldg Distribution Panel, and identifies where Engine Hourmeter should be logged.		

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	5.1.8	PLACE UNLOADER VALVE selector switch in START position.	Operator locates UNLOADER VALVE selector switch, and ensures switch in START position. Note: UNLOADER VALVE selector switch is normally in Start when engine is shutdown.		
*	5.1.9	PRESS AND HOLD BY-PASS VALVE pushbutton.	Operators locates BY-PASS VALVE pushbutton in control panel, and simulates depressing AND holding depressed.		

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*	5.1.10	PLACE SBO-IGN-SWT, ENGINE IGNITION SWITCH in START position until engine starts, then RELEASE to RUN position.	While keeping the Bypass Valve pushbutton depressed, operator simulates placing the SBO-IGN-SWT SBO Air Comp Inst Pnl Engine Ignition Switch in START. Cue: Engine has started. After cue is given, operator releases SBO-IGN-SWT SBO Air Comp Inst Pnl Engine Ignition Switch to RUN. Note: If operator also releases the BYPASS VALVE pushbutton at this point, without checking oil pressure >15 psig, then CUE that the engine has stopped.		
*	5.1.11	When engine oil pressure is >15 psig, RELEASE BY-PASS VALVE pushbutton.	Operator identifies oil pressure reading on gauge labeled oil pressure gauge inside control panel. Cue: Oil pressure is 40 psig. Operator releases BY-PASS VALVE pushbutton.		

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	5.1.12	ALLOW engine to run for 5 minutes to warm up.	Cue: 5 minutes has elapsed.		
*	5.1.13	OPEN 1CA1913 SBO COMPRESSOR DISCHARGE VALVE, to pressurize Control Air header.	Operator locates and describes opening 1CA1913 SBO COMPRESSOR DISCHARGE VALVE Cue: 1CA1913 SBO COMPRESSOR DISCHARGE VALVE, is open.		
*	5.1.14	OPEN 1CA1886, BLACKOUT AIR COMPRESSOR ISOLATION.	Operator locates and describes opening 1CA1886, BLACKOUT AIR COMPRESSOR ISOLATION. Note: 1CA1886 is located near the entrance to Service Building on right side past the U2 mixing bottle Cue: 1CA1886, BLACKOUT AIR COMPRESSOR ISOLATION is open.		

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*	5.1.15	OPEN 2CA584, YARD CONTROL AIR SUPPLY VALVE, to pressurize Aux. Building 1A and 2A Control Air headers.	Operator locates and describes opening 2CA584, YARD CONTROL AIR SUPPLY VALVE. Note: 2CA584 is located near the entrance to Service Building on right side past the U2 mixing bottle. Cue: 2CA584, YARD CONTROL AIR SUPPLY VALVE is open.		
*	5.1.16	PLACE UNLOADER VALVE selector switch in RUN position.	Operator returns to SBO Compressor Building and points out UNLOADER VALVE selector switch describes placing it in RUN position. Cue: JPM is complete.		

JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- _____ 1. Task description and number, JPM description and number are identified.
- _____ 2. Knowledge and Abilities (K/A) references are included.
- _____ 3. Performance location specified. (in-plant, control room, or simulator)
- _____ 4. Initial setup conditions are identified.
- _____ 5. Initiating and terminating Cues are properly identified.
- _____ 6. Task standards identified and verified by SME review.
- _____ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- _____ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. _____ Date _____
- _____ 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: _____

Date: _____

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